Cutaneous Anthrax

Spores are infectious particles that revert to active bacillary form in host tissues. *Bacillus anthracis* is a natural pathogen of livestock, particularly sheep, goats, and cattle. Most human cases are occupationally related, occurring after exposure to infected animals or animal products.

Clinical Features

Cutaneous anthrax is the most common form of the disease and is associated with the lowest morbidity compared to inhalational and gastrointestinal anthrax, which are more virulent and frequently fatal.

Cutaneous Lesions

Cutaneous anthrax occurs when spores enter minor breaks in the skin, typically on exposed areas such as the hands, face, or legs. The infection begins as a painless papule at the site of inoculation, often resembling an insect bite. Over several days, the lesion progresses through stages:

- **Papular → Vesicular → Hemorrhagic bulla with central necrosis → Eschar formation**

The lesion enlarges and becomes surrounded by marked, brawny, non-pitting edema ("malignant edema"). A glistening pseudobulla develops, which becomes hemorrhagic and may appear umbilicated. Central necrosis leads to a characteristic jet-black eschar.

- The ulcer is typically **painless**, a key feature distinguishing it from conditions like brown recluse spider bites.
- Satellite papules or vesicles may appear, sometimes spreading along lymphatics in a sporotrichoid pattern.

- True pustules are rare; the term "malignant pustule" is a misnomer, as the lesion does not suppurate.

One or more lesions may develop depending on the inoculation route. Regional lymphadenitis, fever, malaise, and chills are common, potentially leading to an ulcero-glandular syndrome.

Lesion progression is driven by bacterial toxins and continues despite antibiotic therapy. The eschar dries and separates within 1 to 2 weeks if untreated, but appropriate treatment prevents systemic complications.

Systemic Findings

Cutaneous anthrax may be accompanied by: - Fever - Tachycardia - Hypotension - Tender regional lymphadenopathy

Histopathology

Key histopathologic findings include: - Hemorrhagic edema - Dilated lymphatics - Epidermal necrosis - Sparse inflammatory infiltrate (due to toxin-mediated immune suppression) - Bacilli may be visible in the eschar - Immunohistochemical stains can aid diagnosis

Treatment

- **Naturally occurring anthrax**: Treated with **penicillin** or **doxycycline**.
- **Bioterrorism-associated (weaponized) anthrax**: Initial treatment should include a **fluoroquinolone** (e.g., ciprofloxacin, levofloxacin) due to potential resistance to penicillin and tetracyclines. This recommendation applies even to pregnant women and children. - Once susceptibility is confirmed, therapy may be adjusted accordingly.

Note: Antibiotics kill vegetative bacilli but do not neutralize toxins already produced. Therefore, tissue damage may progress despite effective antimicrobial treatment.

Important Notes

- *Bacillus anthracis* is classified by the Centers for Disease Control and Prevention (CDC) as a **Category A bioterrorism agent** due to its potential for aerosolization.
- There is **no human-to-human transmission** of anthrax.